GARSEF
GREATER AUSTIN REGIONAL SCIENCE AND ENGINEERING FAIR
A Program of the Austin Science Education Foundation

PROJECT GUIDELINES AND TIME REQUIREMENTS

It is vital to create a reasonable timeline with specific goals and due dates. All projects will have to be completed in the STEM Wizard system by the deadline in January. Student Research at middle and high school level is a very involved process, requiring a lot of preparation before students are even allowed to begin experimentation. This is a general guideline for the process.

Review the Fair Schedule for specific dates and deadlines.

Early September (1 week)
Teachers -

- Attend Teacher Orientation

Students/Teacher -

- Choose a Topic: This is often the most difficult process. Students will need guidance. Students should be given assignments to be discussed in class (see "**Topics for Discussion**" attachment). The trick is to forget about science for a moment, try to discover each student's passion and interests. Students doing research on a topic they enjoy will be more dedicated and will make the process more rewarding. Once they have identified a hobby or an interest, it is easy to find the science involved (science is in everything).

Mid September (2 Weeks)
Teachers -

- School and Teacher Registration in STEM Wizard.
- Start student account creation in STEM Wizard.
- Plan and schedule your Science Fair (optional) - ideally before mid-January.

Students -

- Complete background research on selected topics. This is important! Learn as much as possible about the topic and all science related to it. That way you won't repeat someone else's experiment or mistakes. Become an expert in the topic before experimentation.
- Make a copy and start filling out the Student Research Plan Template.
- Formulate a specific question, hypothesis or engineering goal. It should be very narrow, testable, and focused. Start thinking about a procedure and how to test it.
- Write a rationale about why the experiment or engineering goal is important. Explain how the research could affect the real world.

- Think of a good scientific title for the project that is descriptive. A judge should be able to tell what a project is about, just by reading the title. Titles should NOT be creative, they should be DESCRIPTIVE:
    - Bad Example (1): "Magnetism vs. Plants"
    - Bad Example (2): "Got Magnets?"
    - Good Example : "The effect of 100 gauss magnetic field on the growth rate of Phaseolus vulgaris over the period of 10 days"
      In the above "good" example, notice how the title is very specific to what exactly the student is testing:
        - 100 gauss magnet: (typical refrigerator magnet with a specific strength stated).
        - Phaseolus vulgaris: Scientific name of the particular plant the student tested
        - 10 days: The specific time scope of the experiment

Early October (2-3 Weeks)

Students -

- Complete account creation in STEM Wizard by the deadline listed
- Complete your research plan, what you intend to do, what materials you will use and procedure (step by step instructions) as well as how you will analyze the data. This should all be written in future tense, because it is your PLAN and has not been completed yet. Include at least 5 legitimate references (bibliography) which you used when conducting your background research. References should be in **APA format**. They can be web based but should be real peer-reviewed publications. "Google" or "Wikipedia" are not valid references.
- If the research involves human participants, vertebrate animals, microorganisms, hazardous materials/activities/devices, or is a continuation from a previous science fair experiment, students need to fill out the extra information for those sections on the Student Research Plan Template.

Teachers -

- Review all research plans and give students feedback.

Mid October (2 Weeks)

Students -

- Use the rules wizard in STEM Wizard (Senior Division) to determine which forms you will need. Junior Division students should carefully review the MS Safety and Consent Form. Begin filling out those forms and make sure to get the proper signatures. Only actual signatures will be accepted. Do not type names in signature sections of forms. See STEM Wizard Acceptable Signatures document.

Teachers/SRC/IRB -

- Remember to check for ISEF rule violations when reviewing project paperwork. For example, bacteria/mold can NOT be cultured at home.
- Encourage students to make corrections as needed until all projects are approved.

Students -

- Upload any additional required paperwork to STEM Wizard..

Early - Mid November

Students -

- Complete all paperwork, including obtaining required pre-approval signatures.
- Begin the actual experimentation after the project is approved by the teacher and SRC or IRB (if required). The amount of time a student needs will depend upon the study.
- Collect data and take photos.

Teachers -

- Check all student paperwork in STEM Wizard to make sure it is complete and signed before students begin experimentation.
    - Senior Division
        - All students: Forms 1, 1A, 1B, 2A
        - Some students: Forms 2B, 2C, 3, 4, 5A, 5B, 6A, 6B, 7
    - Junior Division
        - All students: Texas MS Safety and Consent Form
        - Some students: Form 2C JR: Middle School Mentor and Research Location
- Monitor any students who are required to do their projects at school.
- Review data analysis, conclusions, and abstracts.

Early January (1-2 Weeks)

Students -

- When complete, analyze and summarize the data and write an Abstract. The Abstract is a brief summary (250 words or less) of the project including:
    - the question posed,
    - hypothesis or engineering goal,
    - general procedure and conclusions.
- If the project required a Form 2C JR (Junior Division) or ISEF Form 2C (Senior Division) - you worked with a mentor, at a research facility or place of business - then have Form 2C filled out and signed by the supervising adult or mentor.
- Submit the abstract in Files and Forms and complete the Additional Information milestone in STEM Wizard.
- Complete any steps left in STEM Wizard.
- Create a project board following the Display and Safety Rules to help present the information to the judges. Remember that GARSEF does not allow demonstration items in the exhibit hall, only boards, binders, laptops/tablets, and project paperwork. Only paper and photos should be on the project board. The board height must be 60" or less.

Teachers -

- Review project boards and all project documentation. Give students feedback.

Mid January (2 weeks)

Teachers -

- Hold a School Science Fair or use other criteria to select your winners and give students a chance to practice getting judged.
- Select the students who will advance to GARSEF by the deadline. DEACTIVATE any students who will not advance to GARSEF.
- Senior Division teachers submit plagiarism reports (abstract and research plan) by the deadline.

Students -

- Students should practice a two minute presentation that explains their project. A good presentation will NOT be a memorized speech. Students should present as if they are teaching the subject to others.
- Participate in the School Science Fair or practice your presentation if there is not a school fair.
- If advancing, make sure all paperwork and forms are completed by the close of STEM Wizard registration.

Late January (GARSEF SRC Review)

Students -

- If advancing from your school, check STEM Wizard for messages from the GARSEF SRC for any paperwork issues or clarification requests.
- Make any changes requested by the GARSEF SRC and resubmit corrected paperwork until you are SRC approved.

Teachers -

- Check with students if they have any messages from GARSEF's SRC. Assist as needed.
- Check with students to make sure they receive approval from GARSEF's SRC.

Early February

Teachers -

- Make sure all GARSEF SRC approved students upload their Submissions for Judges Review by the deadline.
- Communicate with parents and students about check-in, set up, judging, and the award ceremony.

Students -

- Complete all Submissions for Judges Review prior to the deadline.
- Prepare for judging. Students should practice their presentations again. GARSEF Judges will interact, ask questions, etc. A student will have about 10 minutes to present and answer questions from each judge.
- Make plans to attend all of the fair including set up, judging, and the award ceremony.

Topics for Discussion

1. List three of your favorite hobbies or things you enjoy doing.

2. If you had magic powers, list three things you would do to change the world.

3. If you were a great inventor, list three things you might create or make better.

4. What is the most fun thing you have ever done?